

D604

Advanced Analytics

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A data scientist at a 150-person logistics startup stacks five dense layers with no activation functions to model a highly non-linear delivery-time pattern. She trains on GPU instances rented by the hour. What is the core problem?

- A. Five more linear layers are needed to capture the pattern.
- B. Neural networks cannot model non-linear patterns.
- C. The GPUs are too slow.
- D. Without activations, the five layers act as one linear function.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

Training loss on a review-sentiment model at a hotel chain with 90 properties jumps around and becomes NaN after a few batches. The learning rate is 1.0, and the team uses PyTorch. What is the most likely fix?

- A. Raise the learning rate.
- B. Switch to TensorFlow.
- C. Collect reviews from more properties.
- D. Lower the learning rate.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

Over 50 epochs, a defect-detection model at an auto-parts plant shows training loss falling steadily, while validation loss is lowest at epoch 12 and rises after that. The plant runs three shifts. What is the best action?

- A. Train to 100 epochs.
- B. Ignore validation loss, since it always rises eventually.
- C. Use early stopping near epoch 12 and add regularization such as dropout or augmentation.
- D. Collect images from a fourth shift.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Which pair of loss curves indicates underfitting?

- A. Training and validation loss both stay high and close together.
- B. Validation loss below training loss always means underfitting.
- C. Training loss low, validation loss high and rising.
- D. Both losses low and close together.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A retailer with 600 stores wants to classify 224×224 -pixel shelf photos as fully stocked or not. The project sponsor is the VP of store operations. Which model family fits best?

- A. A convolutional neural network (CNN)
- B. A recurrent network (LSTM)
- C. Linear regression on the raw pixel values
- D. k-means clustering of the pixels

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A team must classify the sentiment of customer reviews where word order changes meaning (for example, 'not good' vs. 'good'). Which approach fits?

- A. An image CNN applied to screenshots of the reviews
- B. k-means on review length
- C. A transformer or other sequence model that reads words in context
- D. A bag-of-words model that ignores order

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A veterinary clinic group with 25 clinics has only 800 labeled X-ray images for a fracture classifier. The group recently hired a new CIO. What is the most practical approach?

- A. Give up; deep learning never works with fewer than a million images.
- B. Wait for the new CIO to set a strategy.
- C. Train a deep CNN from scratch.
- D. Fine-tune a pretrained image model (transfer learning).

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A bank's 30-person customer-insights team uses a bag-of-words model that treats 'great' and 'excellent' as unrelated. The reviews are stored in a cloud bucket. Which technique captures that the two words are similar?

- A. A larger vocabulary
- B. Word embeddings
- C. One-hot encoding
- D. Stop-word removal

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A document classifier at a law firm with 12 practice groups must assign each document to exactly one of 5 categories. Documents average 14 pages. Which output layer and loss fit?

- A. 1 linear unit with mean squared error
- B. 5 softmax units with categorical cross-entropy
- C. 5 independent sigmoid units
- D. 1 sigmoid unit with binary cross-entropy

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A network's final layer is a single sigmoid unit trained with binary cross-entropy. What kind of task is it built for?

- A. Predicting a dollar amount
- B. A yes/no classification
- C. Grouping unlabeled images
- D. Choosing one of 10 categories

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A training set has 48,000 images, the batch size is 64, and the model trains for 10 epochs. The team's cloud bill last month was \$2,300. How many iterations (weight updates) happen in each epoch?

- A.** 64
- B.** 7,500
- C.** 750
- D.** 3,072,000

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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12. [TRANSFER — reversed stem / different variable]

A model made 6,000 weight updates over 12 epochs with a batch size of 32. How many training samples were there?

- A.** 16,000
- B.** 2,250
- C.** 192,000
- D.** 500

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A data scientist ~~at a 150-person logistics startup~~ stacks five dense layers with no activation functions to model a highly non-linear delivery-time pattern. ~~She trains on GPU instances rented by the hour.~~ What is the core problem?

Target word: why the network cannot fit the pattern **Decides:** No non-linear activation.

A	Dropped element	More linear layers are still linear.
B	Absolute language	They can, once non-linear activations are added.
C	Camouflage	Hardware speed does not change what the model can represent.
D	Correct	Composing linear maps yields a linear map.

2. Answer D Concept 2

Training loss on a review-sentiment model at a hotel chain ~~with 90 properties~~ jumps around and becomes NaN after a few batches. The learning rate is 1.0, ~~and the team uses PyTorch.~~ What is the most likely fix?

Target word: fix unstable training **Decides:** Learning rate of 1.0 with diverging loss.

A	Inverted operation	Larger steps make divergence worse.
B	Camouflage	The framework does not cause the divergence.
C	Adjacent category	More data does not fix an unstable step size.
D	Correct	Smaller steps stop the loss from overshooting and diverging.

3. Answer C Concept 3

Over 50 epochs, a defect-detection model at an auto-parts plant shows training loss falling steadily, while validation loss is lowest at epoch 12 and rises after that. ~~The plant runs three shifts.~~ What is the best action?

Target word: respond to the loss curves **Decides:** Validation loss turns up after epoch 12.

A	Inverted operation	Longer training deepens the overfitting.
B	Absolute language	Its rise is the overfitting signal.
C	Correct	Stops before the model memorizes training data.
D	Camouflage	Shift count does not diagnose the curves.

4. Answer A Concept 3 · transfer

Which pair of loss curves indicates underfitting?

Target word: reverse: pick the underfitting pattern **Decides:** Both losses stay high.

A	Correct	The model has not learned the pattern even on training data.
B	Absolute language	That can happen with dropout and is not a rule.
C	Inverted operation	That is overfitting.
D	Adjacent category	That is a good fit.

5. Answer A Concept 4

A retailer with 600 stores wants to classify 224 × 224-pixel shelf photos as fully stocked or not. The project sponsor is the VP of store operations. Which model family fits best?

Target word: architecture for images **Decides:** The input is images.

A	Correct	Convolutions capture spatial patterns in images.
B	Confusion set	Designed for ordered sequences such as text.
C	Dropped element	Drops spatial structure and is linear.
D	Adjacent category	Unsupervised; ignores the labels.

6. Answer C Concept 4 • transfer

A team must classify the sentiment of customer reviews where word order changes meaning (for example, 'not good' vs. 'good'). Which approach fits?

Target word: architecture for ordered text **Decides:** Word order matters.

A	Confusion set	Treats text as pictures.
B	Adjacent category	Unsupervised and ignores the content.
C	Correct	Captures how nearby words change meaning.
D	Dropped element	Loses the 'not' in 'not good'.

7. Answer D Concept 5

A veterinary clinic group with 25 clinics has only 800 labeled X-ray images for a fracture classifier. The group recently hired a new CIO. What is the most practical approach?

Target word: approach with little data **Decides:** Only 800 labeled images.

A	Absolute language	Transfer learning works well with small sets.
B	Camouflage	Leadership change does not decide the method.
C	Adjacent category	800 images are too few to learn features from scratch.
D	Correct	Reuses features learned on large datasets.

8. Answer B Concept 6

A bank's 30-person customer-insights team uses a bag-of-words model that treats 'great' and 'excellent' as unrelated. The reviews are stored in a cloud bucket. Which technique captures that the two words are similar?

Target word: represent word similarity **Decides:** Need similar meanings to sit close together.

A	Dropped element	More words without similarity still treats them as unrelated.
B	Correct	Dense vectors place similar words near each other.
C	Confusion set	Every word is still equally unrelated.
D	Adjacent category	Removes common words; does not relate synonyms.

9. Answer B Concept 7

A document classifier at a law firm with 12 practice groups must assign each document to exactly one of 5 categories. Documents average 14 pages. Which output layer and loss fit?

Target word: output and loss **Decides:** Exactly one of 5 classes.

A	Confusion set	For predicting a number.
B	Correct	Softmax gives one probability distribution across mutually exclusive classes.
C	Adjacent category	Fits multi-label tasks where several classes can be true.
D	Dropped element	Handles only two classes.

10. Answer B Concept 7 · transfer

A network's final layer is a single sigmoid unit trained with binary cross-entropy. What kind of task is it built for?

Target word: reverse: from output layer to task **Decides:** One sigmoid output = one yes/no probability.

A	Confusion set	Needs a linear output.
B	Correct	The sigmoid outputs one probability for the positive class.
C	Inverted operation	Unsupervised; there is no label to compute cross-entropy against.
D	Adjacent category	Needs 10 softmax outputs.

11. Answer C Concept 8

A training set has 48,000 images, the batch size is 64, and the model trains for 10 epochs. The team's cloud bill last month was \$2,300. How many iterations (weight updates) happen in each epoch?

Target word: iterations per epoch **Decides:** Samples / batch size.

A	Wrong stem number	The batch size.
B	Adjacent category	Total updates across all 10 epochs.
C	Correct	$48,000 / 64 = 750$.
D	Inverted operation	Multiplies samples by batch size.

12. Answer A Concept 8 · transfer

A model made 6,000 weight updates over 12 epochs with a batch size of 32. How many training samples were there?

Target word: solve for sample count **Decides:** Updates per epoch = 500.

A	Correct	$6,000 / 12 = 500$ per epoch $\times 32 = 16,000$.
B	Inverted operation	$6,000 \times 12 / 32$ flips the operations.
C	Dropped element	$6,000 \times 32$ ignores that updates span 12 epochs.
D	Adjacent category	Iterations per epoch, not samples.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	10	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Activation functions create non-linearity	Without non-linear activations (such as ReLU), any stack of layers collapses to a single linear function.	Adding more linear layers to capture a non-linear pattern.	A deep network with no activation functions.
2	Learning rate	Too high a learning rate makes loss jump or diverge; too low makes training slow.	Raising the learning rate when loss is unstable.	Loss that swings wildly or becomes NaN.
3	Detecting overfitting from loss curves	Training loss falling while validation loss rises means overfitting; stop early or regularize (dropout, augmentation). Both losses high means underfitting.	Training longer once validation loss turns up.	Two loss curves that diverge.
4	Architecture fits the data	Convolutional networks suit images; sequence models (transformers, RNNs/LSTMs) suit text and other ordered data.	Using an order-blind or non-spatial model for images or word order.	Photos vs. sentences where order matters.
5	Transfer learning	With little labeled data, fine-tune a model pretrained on a large dataset rather than training from scratch.	Training a deep network from scratch on a few hundred examples.	A small labeled set for an image or text task.
6	Word embeddings	Embeddings represent words as dense vectors so words with similar meaning sit close together; one-hot and bag-of-words treat every word as unrelated.	Expecting one-hot or stop-word removal to capture similarity.	Synonyms treated as unrelated.
7	Output layer and loss	One-of-many classes → softmax with categorical cross-entropy; binary → one sigmoid unit with binary cross-entropy; numeric → linear output with MSE.	Pairing the wrong output and loss for the task.	"Exactly one of N categories," "yes/no," or "an amount."
8	Epochs, batches, iterations	Iterations per epoch = training samples / batch size; total updates = iterations per epoch × epochs.	Multiplying by batch size, or mixing per-epoch and total counts.	Sample count, batch size, and number of epochs.

Course scope (public description): Advanced Analytics extends analytics techniques from machine learning to artificial intelligence more broadly, including topics in neural networks, deep learning, and natural language processing. The course discusses approaches to developing these models including PyTorch and TensorFlow. Students learn to apply a combination of techniques to solve complex business challenges including computer vision and sentiment analysis. There is no prerequisite for this course.